

CMPG 325 — COMPUTER NETWORKS

Individual Semester Project

Milestone 1 — Client Design Review

Project ID: CMPG325-2026-129

Client ID: CLI-129 — Boitshoko Digital Media Agency (Rustenburg)

Industry: Media

Assigned Networking Challenge: NAT (inside/outside address translation) — Advanced

Student: Sereo, Kgomotso

Student Number: 42859808

Department of Computer Science and Information Systems, NWU

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1. Client Requirements

1.1 Client Overview

Client: Boitshoko Digital Media Agency, Rustenburg (Client ID: CLI-129). Industry: Media. The agency operates from a single site and requires a converged, segmented network supporting day-to-day media production, client-facing sales, and back-office finance operations, with resilient internet access and secure address translation between the internal network and the public internet.

1.2 Scenario Interpretation & Assumptions

The brief fixes the client, industry, addressing block, design constraint, assigned challenge and change request — these are treated as fixed and non-negotiable. It does not fix department structure, headcounts, device types, or service locations. Per the module's project guidance, the correct response to that gap is not to seek more information from the named organisation, but to make a reasonable, defensible network-engineering assumption, document it, and design against it consistently. The assumptions below are the ones this design relies on; each is a normal, low-risk judgement call for an office network of this size and industry, and each is revisited.

#	Gap in the brief	Assumption made	Justification
A1	Department structure	Six functional departments: Management, Finance, Creative & Design, Client Services & Sales, IT/Network Operations, Reception	Standard structure for a small-to-medium media agency; each department maps to a distinct traffic/security profile, which motivates VLAN segmentation
A2	Headcount per department	5–20 users per department today (53 total); see Section 4.1	Sized to a single-site agency small enough for one border router and one core switch, in line with the brief's single addressing block and single-site framing
A3	Shared services	Internal file server, web/portal server, email server, grouped in a Servers/DMZ VLAN	Typical minimum service set for any office LAN; isolating them in their own VLAN limits the blast radius of a compromised end-user device
A4	Backup path scope	Backup ISP provides failover for the whole router (and therefore Finance) rather than a second, Finance-only router	Meets the constraint literally (Finance never loses internet) with one fewer device and one fewer point of failure than a dedicated Finance router; documented as a deliberate simplification, not an oversight
A5	Printers / peripherals	Not modelled as separate devices in Milestone 1; treated as end hosts on their owning department's VLAN if added in Packet Tracer	Brief does not require print services or a specific location; adding this scope would exceed "solve the assigned problem", per the lecturer's guidance not to expand scope unprompted

1.3 Functional Requirements (derived from the project brief)

- FR1 — Provide full IP connectivity across all operational departments at the Rustenburg site: Management, Finance, Creative & Design, Client Services & Sales, IT/Network Operations, and Reception.
- FR2 — Provide all departments with access to shared internal servers (file storage, internal web/portal, email).
- FR3 — Provide outbound internet access to every department for cloud media transfer, client communication and web-based tools, via a primary ISP connection.
- FR4 — Guarantee that the Finance department's internet access survives a primary ISP outage, via a backup internet path (Design Constraint, Section 8 of the brief).

- FR5 — Translate all internal private addressing (10.49.0.0/16) to public/outside addressing before traffic leaves the network (Assigned Networking Challenge: NAT).
- FR6 — Absorb a 25% increase in user numbers in every department without renumbering any existing host or subnet (Client Change Request CR5).

1.4 Non-Functional / Technical Requirements

- Implemented and simulated entirely in Cisco Packet Tracer.
- VLAN segmentation per department to control broadcast domains and to apply department-appropriate access policy.
- Inter-VLAN routing via a router-on-a-stick border router (single physical router, 802.1Q sub-interfaces).
- NAT overload (PAT) on the primary ISP path so the whole 10.49.0.0/16 inside network shares a small outside address pool.
- A second, independent outside path (separate ISP, separate outside network) that keeps Finance reachable if the primary path fails.
- A reproducible, testable .pkt file plus a GitHub portfolio documenting requirements, design, configuration, testing and reflection.

1.5 Design Constraint

Backup internet path required for the finance function. Interpreted as: the router must present two distinct outside paths to the internet (primary and backup ISP), with automatic failover, so that the Finance VLAN's default route — and by extension every VLAN routed through the same border router — is not dependent on a single ISP. See Assumption A4.

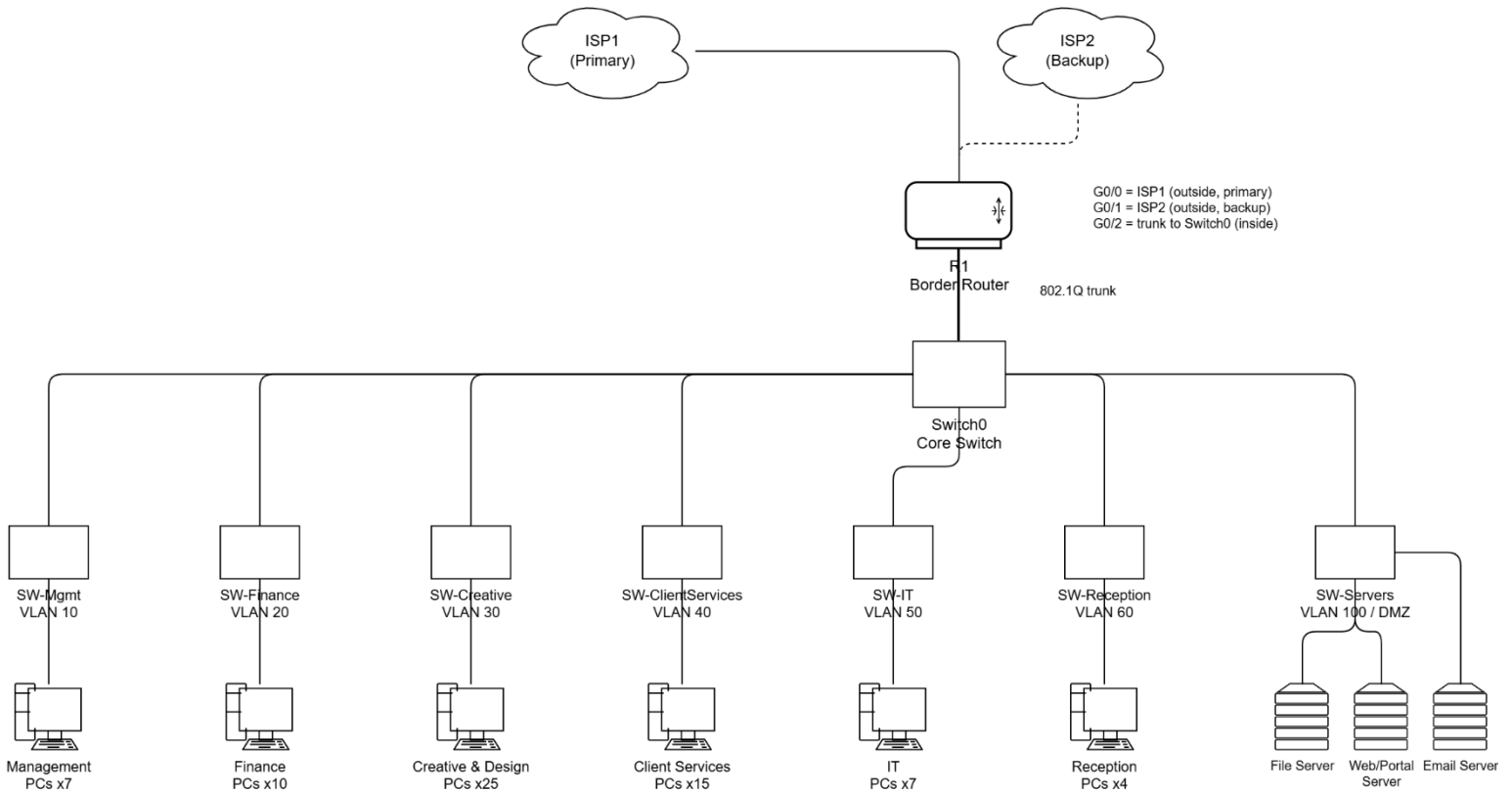
1.6 Client Change Request — CR5

User numbers grow by 25% — the addressing plan must absorb this without renumbering. Interpreted as: every department subnet in Section 4 is sized using the department's headcount after a 25% increase is already applied, and the full 10.49.0.0/16 block is only lightly used (a single /24) so additional departments, sites, or larger-than-expected growth can be added from unused address space without touching any subnet already in service.

2. Physical Topology

Single-site star topology: the border router (R1) connects upstream to two Packet Tracer Cloud objects representing the primary and backup ISPs, and downstream to a core switch. The core switch trunks to seven access switches — one per department plus one for the server/DMZ segment — each carrying its own VLAN and hosting that department's end devices.

Boitshoko Digital Media Agency — Physical Topology



2.1 Device Inventory

Device / Group	Type	Role
R1	1x Cisco 4321 (or 2911) Router	Border router: NAT, inter-VLAN routing (router-on-a-stick), dual WAN
Switch0	1x Cisco 2960 Switch	Core switch — 802.1Q trunk to R1 and to all access switches
SW-Mgmt / SW-Finance / SW-Creative / SW-ClientServices / SW-IT / SW-Reception	6x Cisco 2960 Switch	Department access switches — one per VLAN/department
SW-Servers	1x Cisco 2960 Switch	Server/DMZ access switch — VLAN 100
Servers	File Server, Web/Portal Server, Email Server (Packet Tracer Server devices)	Shared internal services, VLAN 100
End devices	PCs/laptops per department (see Section 4 headcounts)	Department end-user devices
Cloud objects	ISP1 (Primary), ISP2 (Backup)	Packet Tracer Cloud objects simulating the two ISP connections

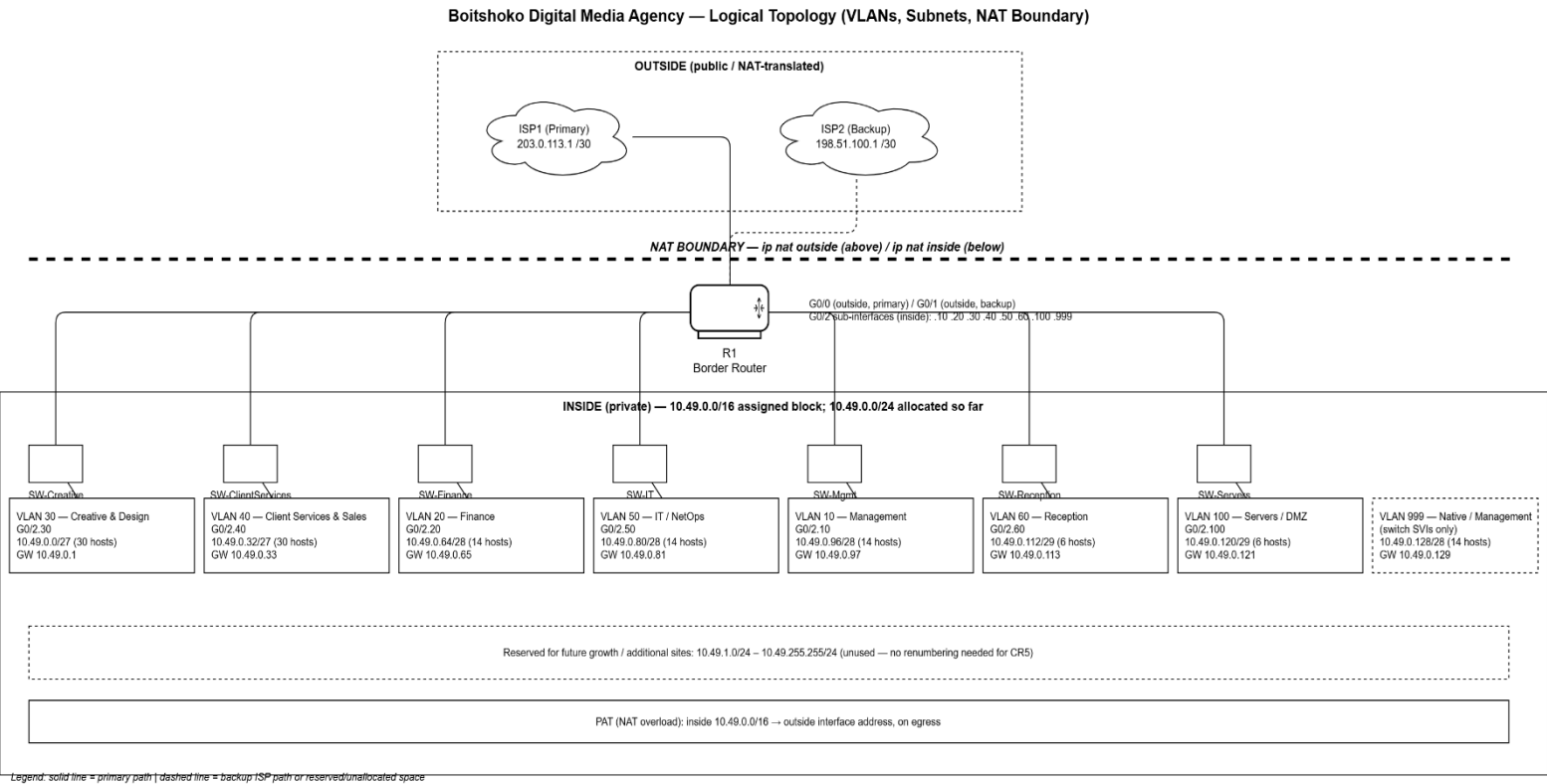
2.2 Physical Connections (summary)

- ISP1 Cloud — R1 G0/0 (primary WAN link)
- ISP2 Cloud — R1 G0/1 (backup WAN link, serves Finance failover)

- R1 G0/2 — Switch0 (trunk, carries all department + server VLANs)
- Switch0 — SW-Mgmt, SW-Finance, SW-Creative, SW-ClientServices, SW-IT, SW-Reception, SW-Servers (trunk uplinks)
- Each access switch — its department's end devices (access ports) or servers (SW-Servers)

3. Logical Topology

The logical view groups the same physical devices by VLAN and IP subnet, and marks the NAT inside/outside boundary: everything inside the router's LAN sub-interfaces uses the private 10.49.0.0/16 block (inside); the two WAN interfaces face the ISPs on public/outside address space and are where NAT overload (PAT) is applied on egress.



3.1 VLAN Plan

VLAN	Department	Subnet	Router Sub-interface
10	Management	10.49.0.96/28	G0/2.10
20	Finance	10.49.0.64/28	G0/2.20
30	Creative & Design	10.49.0.0/27	G0/2.30
40	Client Services & Sales	10.49.0.32/27	G0/2.40
50	IT / Network Operations	10.49.0.80/28	G0/2.50
60	Reception	10.49.0.112/29	G0/2.60
100	Servers / DMZ	10.49.0.120/29	G0/2.100
999	Native / Management (switch SVIs)	10.49.0.128/28	G0/2.999

3.2 NAT Inside/Outside Boundary

- Inside (private): all router sub-interfaces on G0/2 (VLANs 10, 20, 30, 40, 50, 60, 100) — configured as ip nat inside.
- Outside (public): G0/0 toward ISP1 (203.0.113.2/30) and G0/1 toward ISP2 (198.51.100.2/30) — both configured as ip nat outside.
- Primary path: PAT (NAT overload) on G0/0, translating the whole 10.49.0.0/16 inside range to the G0/0 outside address.
- Backup path: a mirrored PAT/NAT pool bound to G0/1, activated by a floating static default route with a higher administrative distance than the primary default route, so traffic — including Finance — fails over automatically if ISP1 goes down.

4. IP Addressing Plan

VLSM applied within the assigned 10.49.0.0/16 block. Every subnet below is sized to the department's headcount after CR5's 25% growth is already added, so the existing addressing does not need to change if that growth occurs. Only 10.49.0.0/24 — a single /24 out of the full /16 — has been allocated; the remaining 255 /24-equivalent blocks (10.49.1.0 – 10.49.255.255) are held in reserve for growth beyond CR5, new departments, or additional sites, again without renumbering anything issued here.

4.1 Inside Addressing (10.49.0.0/24, VLSM)

Department	Current Users	Users +25% (CR5)	Network / Mask	Usable Hosts	Usable Range	Broadcast
Management	5	7	10.49.0.96/28	14	10.49.0.97 – 10.49.0.110	10.49.0.111
Finance	8	10	10.49.0.64/28	14	10.49.0.65 – 10.49.0.78	10.49.0.79
Creative & Design	20	25	10.49.0.0/27	30	10.49.0.1 – 10.49.0.30	10.49.0.31
Client Services & Sales	12	15	10.49.0.32/27	30	10.49.0.33 – 10.49.0.62	10.49.0.63
IT / Network Operations	5	7	10.49.0.80/28	14	10.49.0.81 – 10.49.0.94	10.49.0.95
Reception	3	4	10.49.0.112/29	6	10.49.0.113 – 10.49.0.118	10.49.0.119
Servers / DMZ	4 devices	4 devices	10.49.0.120/29	6	10.49.0.121 – 10.49.0.126	10.49.0.127
Mgmt/Native (switch SVIs)	—	reserve 14	10.49.0.128/28	14	10.49.0.129 – 10.49.0.142	10.49.0.143

4.2 Outside (Public/NAT) Addressing

Documentation-range addressing (RFC 5737 / TEST-NET-3 and TEST-NET-2) is used to simulate ISP-assigned public IPs for the Packet Tracer build, since real public IP space cannot be allocated in a lab environment.

Link	Network	ISP-side IP	Router-side IP
Primary WAN (R1 G0/0 – ISP1)	203.0.113.0/30	203.0.113.1 (ISP1)	203.0.113.2 (R1)
Backup WAN (R1 G0/1 – ISP2)	198.51.100.0/30	198.51.100.1 (ISP2)	198.51.100.2 (R1)

4.3 CR5 Growth Absorption — Design Rationale

- Every subnet size already includes the +25% growth column, so activating CR5 requires zero re-addressing of existing hosts.

- Subnets were placed contiguously from 10.49.0.0 upward using VLSM (largest-first), leaving the entire 10.49.1.0/24 – 10.49.255.255/24 range untouched.
- If any single department grows beyond the figure already provisioned here, the next unused block in the reserved range can be allocated to it (or DHCP scope simply widened within the existing /27 or /28 where slack remains) without renumbering any other department.
- The backup WAN and primary WAN are on separate /30 outside networks, independent of the inside addressing, so ISP changes never require inside renumbering either.

5. Initial GitHub Repository

An initial repository has been structured to hold the full portfolio of evidence for the project, following the brief's requirement for a professional README and organised evidence covering client requirements, network design, IP addressing, topology.

5.1 Repository Structure

- README.md — project summary, scope, milestones (provided as a separate file alongside this document).
- docs/ — client-requirements.md, design-decisions.md, reflection.md.
- topology/ — physical-topology.drawio, logical-topology.drawio.
- ip-addressing/ — ip-addressing-plan.md (source for Section 4 above).

5.2 Milestone 1 Commit Plan

- Commit 1 — Initialise repository, add README.md and folder structure.
- Commit 2 — Add docs/01-client-requirements.md (Section 1 content).
- Commit 3 — Add topology/physical-topology.drawio and logical-topology.drawio.
- Commit 4 — Add ip-addressing/ip-addressing-plan.md (Section 4 tables).
- Commit 5 — Tag milestone-1 once all four artefacts are pushed.